

Bezeichnung / Funktionsblock	Übertragungsverhalten		Mathematische Funktionen		Grafische Darstellungen	
	statisch	dynamisch	Zeitbereich	Frequenzbereich	Ortskurve	Bodediagramm
P-Glied 			$x_a(t) = K_P \cdot x_e(t)$			
P-T₁-Glied 			$T_1 \cdot \dot{x}_a(t) + x_a(t) = K_P \cdot x_e(t)$ Sprungantwort: $x_a(t) = K_P \cdot x_{e0} \cdot \left(1 - e^{-\frac{t}{T_1}}\right)$			
P-T₂-Glied 			$T_1 T_2 \cdot \ddot{x}_a(t) + (T_1 + T_2) \cdot \dot{x}_a(t) + x_a(t) = K_P \cdot x_e(t)$			
P-T₂S-Glied 			$\frac{1}{\omega_0^2} \cdot \ddot{x}_a(t) + 2 \frac{D}{\omega_0} \cdot \dot{x}_a(t) + x_a(t) = K_P \cdot x_e(t)$			

Grundglieder

F Frequenzgang (komplex)
 ω Kreisfrequenz [rad/s]

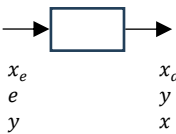
$\Re(F)$ Realteil
 $\Im(F)$ Imaginärteil

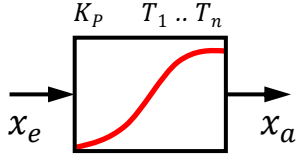
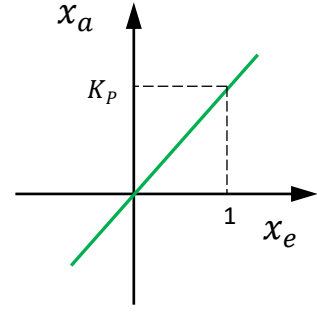
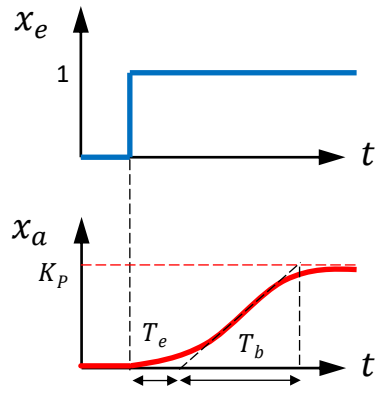
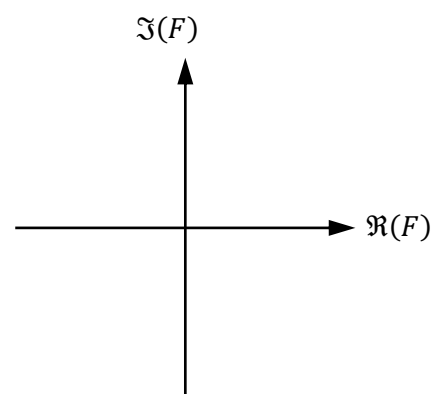
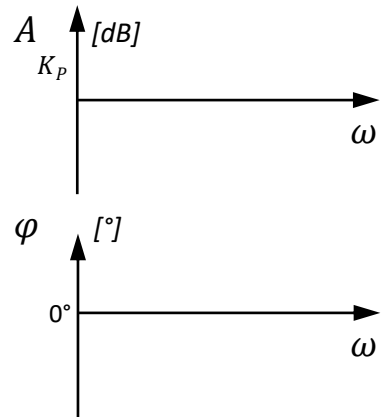
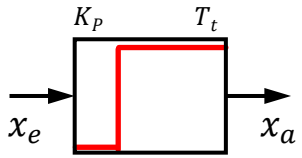
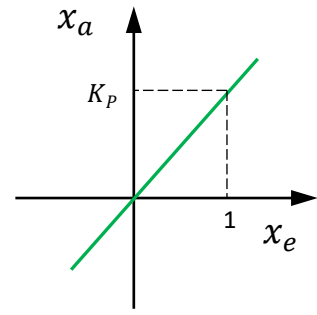
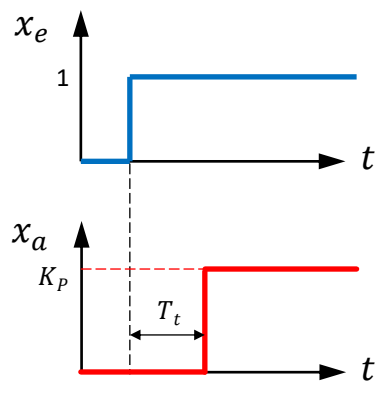
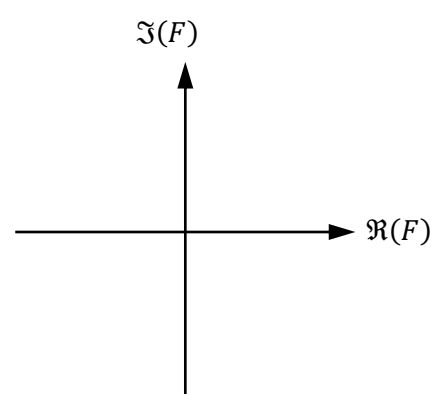
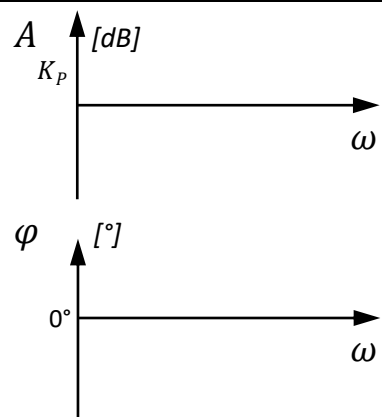
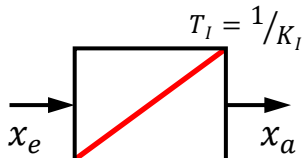
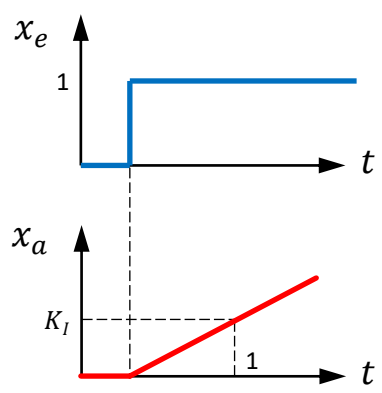
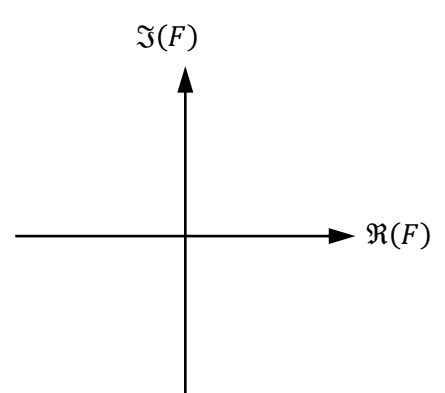
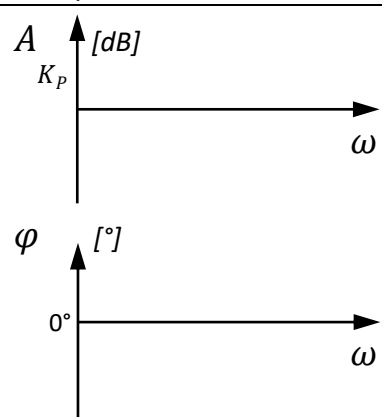
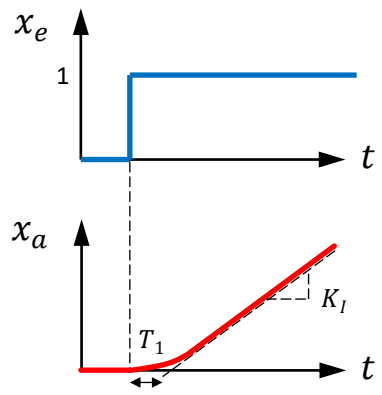
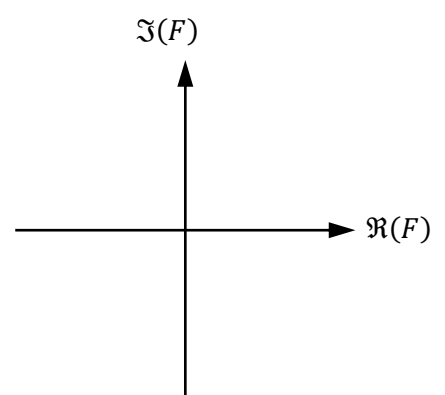
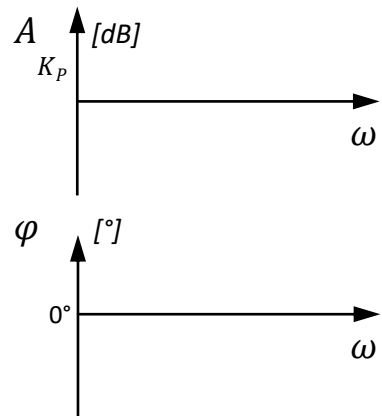
$A = |F|$ Amplitudengang (Betrag) [dB]
 $\varphi = \angle F$ Phasengang (Phasenverschiebung) [°]

$$F(j\omega) = \frac{b_m \cdot (j\omega)^m + \dots + b_2 \cdot (j\omega)^2 + b_1 \cdot j\omega + b_0}{a_n \cdot (j\omega)^n + \dots + a_2 \cdot (j\omega)^2 + a_1 \cdot j\omega + a_0}$$

$$A_{dB} = 20 \cdot \log_{10}(A)$$

$$A = 10^{(A_{dB}/20)}$$



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P-T_n-Glieder  In BORIS: $T_1 = T_2 = \dots = T_n$			-			
P-T_t-Glieder (Totzeit) 			$x_a(t) = K_P \cdot x_e(t - T_t)$			
I-Glieder 	-		$x_a(t) = K_I \cdot \int x_e(t) dt$ $= \frac{1}{T_I} \cdot \int x_e(t) dt$			
I-T₁-Glieder Dieses Glied muss in BORIS zusammengesetzt werden aus I-Glied und P-T ₁ -Glieder (analog für I-T ₂ , I-T _t etc.)	-		$T_1 \cdot \dot{x}_a(t) + x_a(t) = K_I \int x_e(t) dt$			

Grundglieder

F Frequenzgang (komplex)
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$\Re(F)$ Realteil
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$A = |F|$ Amplitudengang (Betrag) [dB]
 $\varphi = \angle F$ Phasengang (Phasenverschiebung) [°]

$$F(j\omega) = \frac{b_m \cdot (j\omega)^m + \dots + b_2 \cdot (j\omega)^2 + b_1 \cdot j\omega + b_0}{a_n \cdot (j\omega)^n + \dots + a_2 \cdot (j\omega)^2 + a_1 \cdot j\omega + a_0}$$

$A_{dB} = 20 \cdot \log_{10}(A)$
 $A = 10^{(A_{dB}/20)}$

Allgemein:
Regler (RE):
Regelstrecke (RS):

x_e

e

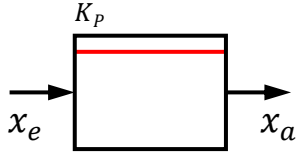
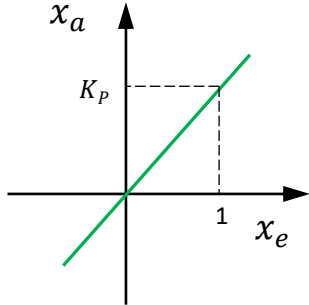
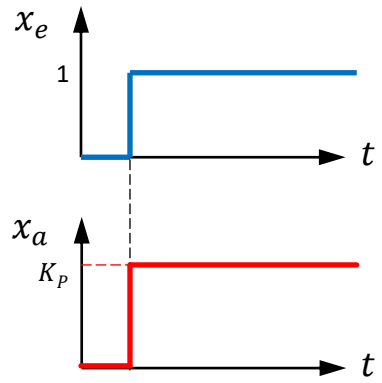
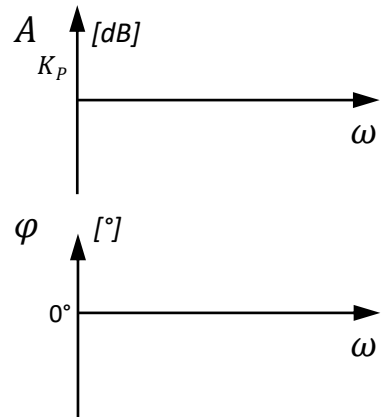
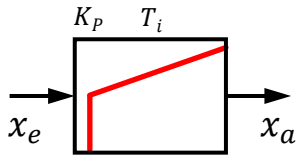
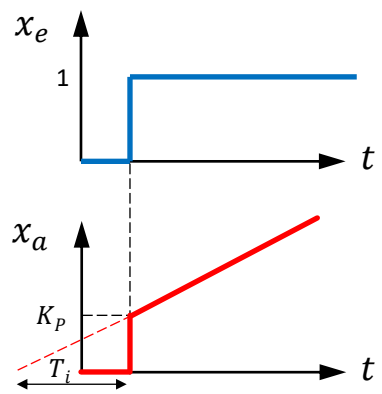
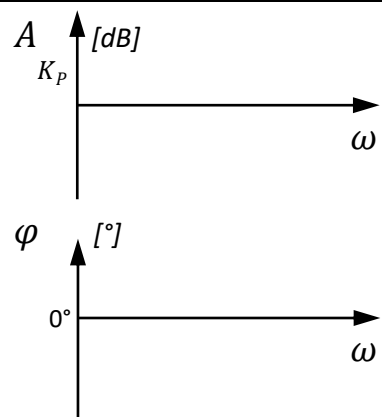
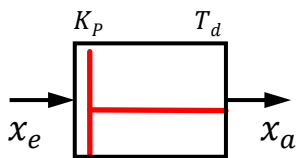
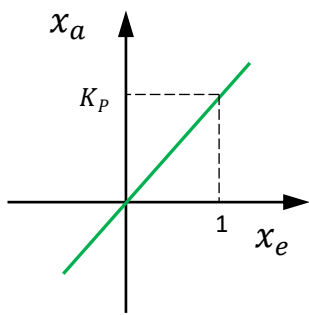
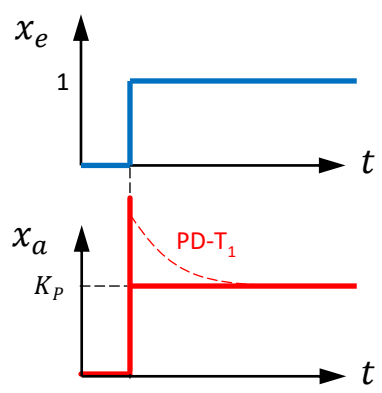
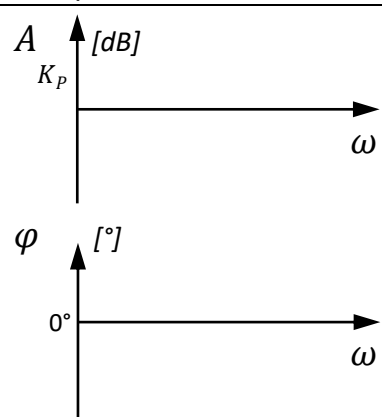
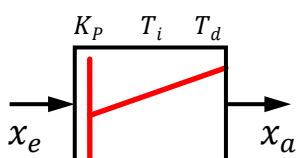
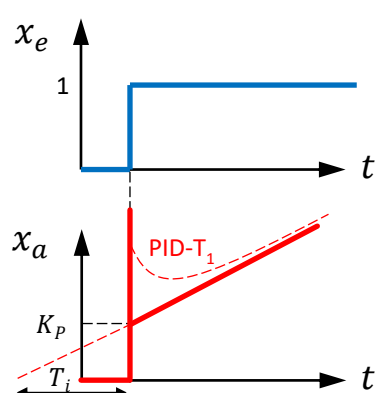
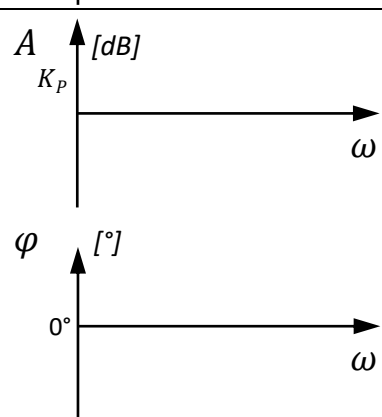
y

x_a

y

x

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D-T₁-Glied			$T_1 \cdot \dot{x}_a(t) + x_a(t) = K_D \cdot \dot{x}_e(t)$ Sprungantwort: $x_a(t) = \frac{K_D}{T_1} \cdot e^{\frac{-t}{T_1}}$			

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P-Regler (= P-Glied)  (analog: I-Regler = I-Glied)			$y(t) = K_P \cdot e(t)$			
PI-Regler  (PID-Block mit deaktiviertem D-Anteil)	-		$y(t) = K_P \cdot \left(e(t) + \frac{1}{T_i} \int e(t) dt \right)$			
PD-Regler  (PID-Block mit deaktiviertem I-Anteil)			$y(t) = K_P \cdot (e(t) + T_d \cdot \dot{e}(t))$			
PID-Regler 	-		$y(t) = K_P \cdot \left(e(t) + \frac{1}{T_i} \int e(t) dt + T_d \cdot \dot{e}(t) \right)$			

Grundglieder